

Radar Search Pipeline AS IS

Current implementation guide for Radar candidate and signal search

Status: AS IS

Product area: Radar candidate and signal search

Updated after slice: 0.7.6.3.6.1

Last updated: 2026-06-27

Canonical source: current implementation, tests, ROADMAP.md, and Radar run diagnostics

Generated PDF: docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.pdf

1. Purpose

This document explains the current Radar search pipeline as it is implemented today. It is the operational map for candidate discovery, source handling, identity enrichment, signal search, diagnostic states, and benchmark evaluation.

The document has two jobs:

- help a user or new developer understand how Radar search currently works;
- help agents and developers find the correct extension point before changing planner, retrieval, extraction, registry lookup, checkpoints, signal search, dossier projection, or evaluation.

The Markdown file is the source of truth. The PDF is a generated review artifact. Mermaid source can remain in Markdown, but the PDF must contain rendered diagrams instead of raw diagram notation.

2. AS IS And TO BE Rule

There is exactly one current AS IS document:

- docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.md
- docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.pdf

Substantial Radar pipeline changes must start with a TO BE design:

- docs/radar/to-be/RADAR_SEARCH_PIPELINE_TO_BE_<slice>.md

The TO BE document describes the intended algorithmic change before implementation. After implementation and validation, the accepted behavior is merged into this AS IS document and the PDF is regenerated.

3. Glossary

Term	Meaning
Radar definition	Persisted active configuration for one Radar, including criteria, signals, source policy, and scoring settings.
Candidate universe	The broad set of source-backed entities known to the run, including legal entities and review-needed sites, branches, projects, or assets.

Term	Meaning
Product candidate	A strict account candidate shown as a scored product row. Product candidates should be legal entities or resolved account-level entities.
Review-needed entity	A source-backed entity retained upstream with review flags, but not promoted as a confident product account.
Signal	A monitored buying-intent or activity indicator evaluated for candidate entities.
Source	A user-selected information source in Radar settings.
Connector profile	Human-readable config describing a connector, good inputs, bad inputs, expected facts, limitations, and credentials.
Capability card	Backend-compiled machine-readable connector capability used by preflight, planner input, and execution guards.
Planner source card	Compact planner-facing source description compiled from capability, source definition, and source obligation.
Source obligation	User-facing source usage mode such as required_for_identity, required_for_coverage, preferred, fallback, or disabled.
Retrieved source	URL/snippet/citation/source material returned by retrieval before extraction.
Analyzed source	Source material inspected by extraction or diagnostics but not necessarily linked to evidence.
Used source	Evidence-bearing source linked to a product candidate or finding.
Evidence ref	Stable source reference used to link extracted candidate/finding rows to normalized source records.
Checkpoint	Application-owned review point that decides whether execution continues, retries, expands, revises, stops for review, or fails hard.
Execution budget	Radar task budget for discovery, gate, signal, provider, and run-total work.
External-call budget	Budget for external actions such as OpenRouter calls, DaData lookups, provider retries, and source verification requests.
Budget reserve	A sub-allocation inside the run budget for registry identity, recall expansion, official coverage, open-web coverage, extraction recovery, or signal search.
Expansion target	A prioritized source-backed target that weak discovery should search next, such as a holding, legal entity, production site, branch, alias/language variant, or explicit benchmark target.
not_observed	A searched signal with no evidence found. It must not mean "not searched".
not_searched_*	Explicit unsearched state caused by budget, policy, missing scope, or pending output.

4. High-Level Pipeline

Figure 1. End-to-end Radar execution flow



The pipeline is application-owned. Integrations execute bounded provider tasks, but they do not own scoring, review semantics, source obligations, or final candidate state.

5. Backend Roles

API route

Current owner	src/power_web_os/api
Responsibility	Create runs, expose status, candidates, dossier, runtime/preflight DTOs.
Must not own	Provider calls, scoring, SQL queries in routes.

Worker endpoint

Current owner	src/power_web_os/jobs
Responsibility	Load run id, call application executor, persist status.
Must not own	Provider normalization or domain decisions.

Workflow wrapper

Current owner	src/power_web_os/workflows
Responsibility	Wrap application execution in workflow state when needed.
Must not own	SQLAlchemy queries, provider logic, scoring semantics.

Active definition adapter

Current owner	src/power_web_os/application/live_radar_definition_runtime.py
Responsibility	Convert persisted definition payload into runtime Radar payload.
Must not own	HTTP/provider details.

Connector profile registry

Current owner	src/power_web_os/application/connector_profiles.py and connector_capability_defaults.py
Responsibility	Load connector profiles and compile capability cards.
Must not own	User source obligations or provider calls.

Planner input builder

Current owner	src/power_web_os/application/live_radar_discovery_planning.py
Responsibility	Build source-card-aware planning input and deterministic fallback plans.
Must not own	Final truth or provider execution.

Planner adapter

Current owner	src/power_web_os/integrations/openrouter_discovery_planner.py
Responsibility	Ask OpenRouter for structured discovery plans.
Must not own	Source policy enforcement.

Plan acceptance

Current owner	src/power_web_os/application/live_radar_plan_acceptance.py
Responsibility	Validate capabilities, source obligations, and safe repairs.
Must not own	Provider calls.

Retrieval plan compiler

Current owner	src/power_web_os/application/live_radar_retrieval_plan.py
Responsibility	Convert accepted plan steps into bounded provider task cards.
Must not own	Scoring.

Web retrieval/extraction provider

Current owner	src/power_web_os/integrations/live_radar_openrouter.py
Responsibility	Execute OpenRouter web/retrieval/extraction tasks under budget guard.
Must not own	Execution budgets, final scoring, source obligations.

Source registry/provider orchestration

Current owner	src/power_web_os/application/radar_source_providers.py
Responsibility	Execute structured company registry providers for allowed stages.
Must not own	Signal evidence replacement.

Registry lookup term generator

Current owner	src/power_web_os/application/radar_registry_lookup_terms.py
Responsibility	Build concrete lookup terms for registry providers.
Must not own	Broad web discovery.

Search expansion service

Current owner	src/power_web_os/application/radar_search_expansion.py, radar_search_expansion_models.py, and radar_search_expansion_support.py
Responsibility	Build prioritized expansion target queues and bounded source-profile-driven query variants when discovery/coverage is weak.
Must not own	Direct provider calls.

Search expansion executor

Current owner	src/power_web_os/application/live_radar_search_expansion_execution.py
Responsibility	Execute checkpoint-approved expansion tasks under source policy and reserve budgets.
Must not own	Choosing checkpoint decisions.

Extraction contract/repair

Current owner	src/power_web_os/application/live_radar_extraction_contract.py
Responsibility	Validate and repair provider payload shape when deterministic repair is safe.
Must not own	Silently converting unrecoverable output into success.

Checkpoint service

Current owner	src/power_web_os/application/live_radar_checkpoints.py
Responsibility	Decide continue, retry, expand, repair, revise, stop, or fail.
Must not own	Direct HTTP/provider calls.

Checkpoint action executor

Current owner	src/power_web_os/application/live_radar_checkpoint_actions.py
Responsibility	Apply approved checkpoint actions under budgets and policy.
Must not own	Unbounded loops.

Entity resolution

Current owner	src/power_web_os/application/live_radar_entity_resolution.py
Responsibility	Distinguish legal entity, branch, production site, project, asset, and unknown entity.
Must not own	Provider transport.

Candidate universe support

Current owner	src/power_web_os/application/live_radar_universe.py and retrieved-candidate helpers
Responsibility	Preserve source-backed legal entities and review-needed upstream entities.
Must not own	Product precision claims.

Dossier projection

Current owner	src/power_web_os/api/radar_dossier_mappers.py and related mappers
Responsibility	Explain lifecycle, diagnostics, checkpoints, budgets, candidates, and sources.
Must not own	Mutating run behavior.

Evaluation

Current owner	src/power_web_os/radar_evaluation.py
Responsibility	Compare persisted run/dossier output to curated baseline.
Must not own	Live provider calls.

6. Inputs And Runtime Configuration

The live run is driven by these inputs:

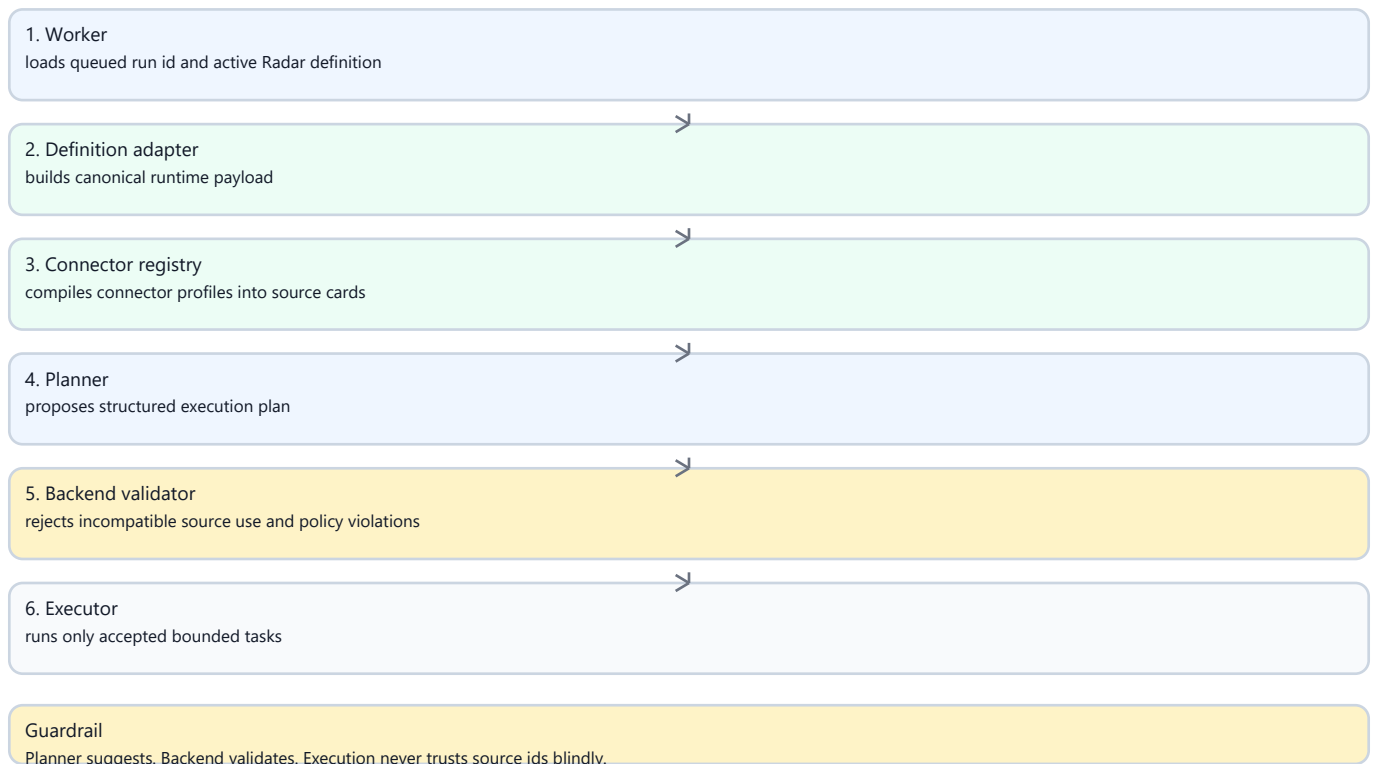
- active RadarDefinitionRecord loaded by radar_id;
- canonical runtime Radar payload built from the active definition;
- global_search_policy.sources;
- source usage obligations from Radar settings;
- connector profiles under config/connectors;
- compiled capability cards and planner source cards;
- runtime config from .env and task context;
- execution budgets and external-call budgets;
- run profile such as live, smoke, benchmark_smoke, or benchmark_live;
- persisted run metadata from API queue time and worker execution time.

Secrets stay in .env or deployment secret storage. They are never passed to

planner source cards, dossier, journal, evaluation reports, or AS IS/TO BE docs.

7. Planning Loop

Figure 2. Planner, source cards, and backend validation



Current planning behavior:

1. The worker loads the active persisted Radar definition. The hardcoded mini Radar is legacy/offline fallback only.
2. Connector profiles are compiled into source cards for configured sources.
3. Planner input includes source cards and source obligations. Source cards say what a source can do; obligations say how strongly the selected source must be used.
4. The planner may use OpenRouter and must return a structured plan.
5. Backend validation rejects incompatible source use, disabled sources, missing required source use, and lookup-only sources used for broad discovery.
6. Invalid plans can enter a bounded revision loop. If revision remains invalid, execution stops as review-needed or policy-blocked instead of falling into blind execution.

Planner calls count against OpenRouter external-call budgets, including total OpenRouter run budget and planner-role budget.

8. Source Cards And Connector Profiles

Connector profiles are human-facing config. They describe a connector in terms a connector author can understand:

- display name;
- description;

- good inputs;
- bad inputs;
- expected facts;
- limitations;
- credential environment variable names;
- runtime provider id.

Application code compiles those profiles into capability cards:

- lookup-only or broad-discovery-capable;
- identity/enrichment/coverage/signal applicability;
- required input kinds;
- returned fact kinds;
- useful-result criteria;
- accepted input shapes;
- bad input shapes;
- non-blocking outcomes;
- language/alias hints;
- connector capability class;
- credential requirements.

Planner source cards are compact, product-safe versions of the capabilities.

They contain no credentials, API keys, headers, tokens, or provider secrets.

9. Retrieval, Extraction, And Recovery Loop

The retrieval/extraction path is split conceptually:

1. Retrieval obtains ranked sources, snippets, URLs, citations, and source outcomes.
2. Extraction maps task cards plus retrieved material into structured observations, candidates, source outcomes, coverage findings, and candidate universe gaps.
3. Schema validation checks provider output shape.
4. Deterministic repair attempts safe shape fixes.
5. If repair fails, one bounded primary retry can be attempted with strict JSON context.
6. If configured, one bounded backup extraction model retry can be attempted for extraction-stage tasks only.
7. If still invalid, the branch stops with an explicit diagnostic reason such as `primary_schema_invalid`, `backup_schema_invalid`, `backup_not_configured`, or `budget_exhausted_before_backup`.

Extraction recovery attempts are recorded in `extraction_recovery_records` and must count against external-call and provider-retry budgets.

10. Registry Lookup Loop

Company registry providers are structured identity/enrichment sources. DaData is the first implementation. Registry providers are not broad discovery engines and must not replace web evidence for signals.

The current lookup flow:

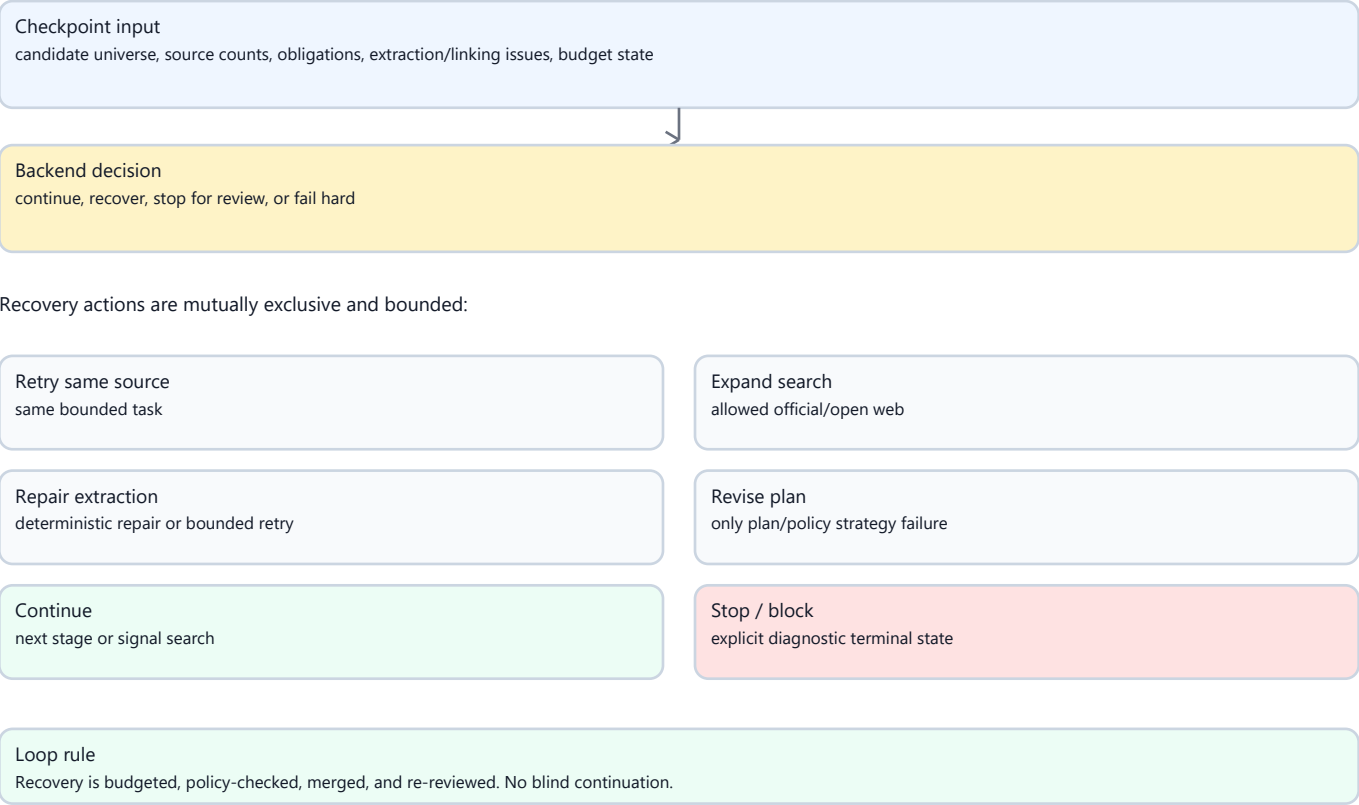
1. Build concrete lookup terms from candidate names, retrieved snippets, identifiers, Russian aliases, English aliases, legal-form variants, and short fragments.
2. Reject placeholders such as "candidates from step 1" and broad natural language discovery queries.
3. Try terms in bounded order: identifiers, Russian legal-form terms, Russian short names, then English aliases.
4. Count each lookup term against DaData lookup budget.
5. Stop after a useful match.
6. Preserve each lookup attempt in `registry_lookup_attempts`.

Important semantics:

- one alias `no_match` is not a hard block;
- `registry_lookup_insufficient` means there was no concrete registry input;
- `identity_not_confirmed_after_all_terms` means concrete terms were tried but identity still was not confirmed;
- ambiguous but source-backed entities can be retained as review-needed upstream entities or linked facts.

11. Search Expansion Loop

Figure 3. Checkpoint decision and recovery loop



When discovery or coverage is weak, RadarSearchExpansionService first builds a prioritized expansion target queue, then compiles bounded query variants for the highest-priority targets.

Checkpoint recovery now uses the same target-aware expansion path. If the run has weak recall, low linked-source coverage, explicit benchmark target hints, or source-backed unresolved gaps, the checkpoint can choose expand_sources before asking for a plan revision. This prevents evidence_linking_failed from automatically consuming the revision cap when the real problem is that important targets were not retrieved yet.

Target classes:

- holding_or_group_target;
- production_site_or_branch_target;
- known_subsidiary_or_legal_entity_target;
- source_backed_universe_gap_target;
- alias_or_language_variant_target;
- benchmark_baseline_like_target only when explicit benchmark context exists;
- low_confidence_registry_suggestion_target.

Each target records target_id, target_label, target_type, source_refs, why_target_exists, priority, allowed source ids, expected fact kinds, budget_reserve_key, execution status, and not-searched reason.

Variant selection is target-aware. The planner no longer takes a flat top-N list

of query variants. It first deduplicates variants, then walks target-type lanes and target ids so one holding/legal-entity target cannot consume all early smoke expansion slots. Production-site and branch targets are lane-prioritized because they drive review recall, while product candidate projection remains strict. Targets that are generated but do not receive an execution slot are kept in `targets_not_searched` with `not_searched_reason=not_selected`.

Query variants are compiled from connector source cards and policy:

- official/domain-capable sources produce `site:<domain> <target>` plus relation and industrial coverage queries;
- broad web sources produce relation, membership, identity, and industrial/site queries;
- lookup-only registry sources are not used for broad expansion tasks;
- source-specific domains come from the selected source/profile, not a hardcoded production branch;
- relation terms come from the radar definition and target context.

Expansion respects source policy and budget reserves. If open web is disabled, open-web variants are not generated. If an official source is disabled, official-domain variants are not generated. If a reserve is exhausted, the target/task is recorded in `targets_not_searched` with `not_searched_reason` instead of disappearing from diagnostics.

Production-site, branch, asset, and project targets use the dedicated `production_site_coverage_probe` reserve. In smoke profile this reserve defaults to 2; `benchmark_smoke` overrides it to 3 so a bounded SIBUR contour smoke can attempt the three curated production-site misses before broader live benchmarking.

Expansion diagnostics include `expansion_target_summary_by_type`, `search_expansion_results_by_target_type`, `targets_not_searched`, and reserve counters. Evaluation can therefore distinguish `expansion_not_selected`, `expansion_budget_limited`, `expansion_searched_no_support`, and `present_not_projected` instead of reporting every production-site miss as a blank `not_retrieved_in_run`.

For benchmark runs only, `task_context.benchmark_target_hints` can seed `benchmark_baseline_like_target` expansion targets. Production runs ignore those hints unless an explicit `benchmark_profile` is present.

12. Candidate Universe And Entity Resolution

The upstream candidate universe is recall-first. It can retain source-backed entities that are not yet strict product accounts.

Current rules:

- legal entities can become product candidates when evidence and resolution are

sufficient;

- weak legal-entity mentions can remain review-needed universe entries;
- branches, factories, production sites, projects, and assets can be retained

as review-needed upstream entities or linked facts;

- unresolved sites, projects, or assets must not become high-confidence product

account candidates;

- review flags explain why an entity needs human attention.

Common review flags include:

- `requires_human_review`;
- `not_standalone_legal_entity`;
- `registry_match_ambiguous`;
- `official_source_cross_checked`;
- `candidate_universe_gap`.

Product candidate projection remains precision-first. Upstream universe retention is allowed to be broader than product account output.

13. Checkpoints And Adaptive Actions

The checkpoint service reviews execution after key stages:

- after discovery;
- after qualification gates;
- after coverage checks;
- before signal search.

Possible decisions:

- `continue`;
- `retry_same_source`;
- `expand_sources` or search expansion;
- `repair_extraction`;
- `retry_extraction`;
- `revise_plan`;
- `stop_review_needed`;
- `fail_hard`.

Actions are bounded:

- no unbounded retries;
- no policy bypass;
- no hidden broad fallback;
- weak recall with uncovered target hints runs target-aware expansion before revision;
- repeated unlinked evidence after expansion can still route to `revise_plan`;
- no signal search until the pre-signal checkpoint allows it;
- all adaptive provider calls count against budgets.

If a stop is diagnostic rather than a runtime failure, the run can still produce

a completed snapshot with `stopped_for_review_reason` and explicit checkpoint metadata.

14. Signal Search

Signal search is not candidate discovery. It runs only after candidate universe and qualification/coverage checkpoints allow it.

Rules:

- signal tasks should use web/official evidence sources, not registry enrichment sources unless a connector capability explicitly supports signal evidence;
- signal prompts include one signal and the current candidate scope;
- signal extraction must not add new scored candidates;
- new entities found during signal search go to candidate universe gaps or diagnostics;
- unsearched signals become `not_searched_*`, not `not_observed`;
- `not_observed` is reserved for searched-negative evidence or invalid searched evidence.

15. Budget Model

Radar has two complementary budget layers.

Execution budgets:

- discovery tasks per rule;
- gate tasks per candidate/rule;
- signal tasks per candidate/signal;
- provider task keys;
- total web tasks per run.

External-call budgets:

- total OpenRouter HTTP calls;
- OpenRouter planner calls;
- OpenRouter web task calls;
- OpenRouter server-tool web searches reported after the call;
- DaData/company registry lookups;
- source verification requests;
- provider retries per task.

Smoke and benchmark profiles set bounded task context budgets so acceptance runs do not spread into uncontrolled provider calls. OpenRouter latency alone is not an error; the controllable unit is number of external actions.

Budget reserves add an intent-level layer under the same run:

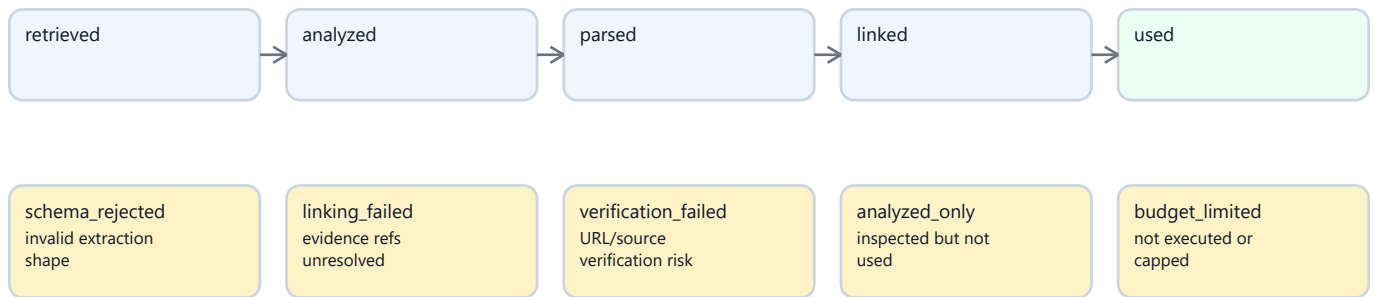
- `primary_discovery`;
- `registry_identity`;
- `recall_expansion`;

- official_coverage_probe;
- open_web_coverage_probe;
- production_site_coverage_probe;
- extraction_recovery;
- signal_search.

Reserve counters are reported as budget_reserve_counters. Exhausted reserves are reported as budget_reserve_exhaustion_events and should produce explicit not-searched diagnostics for skipped expansion targets or registry identity attempts.

16. Source Lifecycle

Figure 4. Source lifecycle and rejection branches



Product source list includes only 'used'. Dossier lifecycle keeps all diagnostic states.

Product source lists remain strict: only evidence-bearing used sources appear as product sources. Dossier/source lifecycle diagnostics are broader and must show retrieved, analyzed, rejected, unlinked, verification-limited, and budget-limited sources.

17. Dossier, Journal, Trace, And Evaluation

Radar exposes several diagnostic surfaces:

- run status and run metadata;
- dossier summary and source lifecycle;
- candidate universe and product candidates;
- checkpoint decisions and adaptive actions;
- source cards and capability validation;
- source obligation decisions and runtime outcomes;
- external-call budget counters;
- extraction recovery records;
- registry lookup terms and attempts;
- expansion target queue;
- search expansion query variants and results grouped by target;
- targets not searched and budget reserve counters;
- registry ambiguity fan-out summary;
- technical trace for sanitized developer inspection;
- benchmark and evaluation reports.

Forbidden everywhere:

- API keys;
- authorization headers;
- bearer tokens;
- raw prompts when not explicitly sanitized;
- raw hidden chain-of-thought;

- raw provider dumps that contain secrets or hidden reasoning keys.

18. Evaluation Loop

The benchmark/evaluation layer reads persisted run output and dossier data. It does not call OpenRouter, DaData, or source providers.

Current SIBUR evaluation channels:

- strict_recall for legal-entity baseline hits;
- review_recall for production-site, branch, asset, or project hits retained

as review-needed universe entities or linked facts;

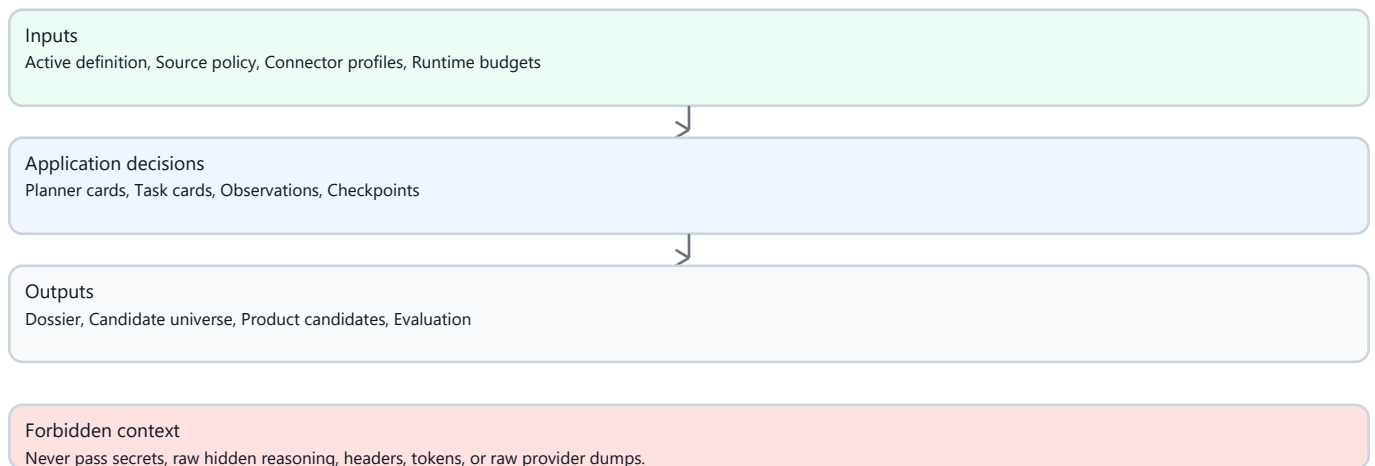
- precision for strict product account candidates;
- false positives;
- false negatives;
- ambiguous matches;
- evidence quality buckets;
- optional coverage probe output that is diagnostic only and does not change the

original metrics.

Evaluation is a measurement layer. If it exposes poor quality, the fix should be planned as a follow-up slice rather than hidden inside the evaluation code.

19. Context Management

Figure 5. Context boundaries across pipeline roles



Planner

Allowed context	Radar goal, source cards, obligations, compact criteria/signals, budget hints.
Forbidden context	Secrets, raw provider dumps, hidden reasoning, credentials.

Plan reviser

Allowed context	Product-safe checkpoint facts and capability validation errors.
Forbidden context	Raw traces, hidden reasoning, secret-bearing payloads.

Web extractor

Allowed context	Task card, retrieved sources, expected schema, current candidate scope.
Forbidden context	Full unrelated run history, secrets, scoring decisions.

Backup extractor

Allowed context	Failed extraction reason and strict task-specific JSON context.
Forbidden context	Planner reasoning, broad source policy mutation.

Registry provider

Allowed context	Concrete legal names, INN, OGRN, legal-form variants, strong aliases.
Forbidden context	Broad discovery queries and placeholders.

Checkpoint service

Allowed context	Counts, outcomes, warnings, budgets, source obligations, extraction issues.
Forbidden context	Direct provider credentials or raw hidden reasoning.

Entity resolver

Allowed context	Provider observations, registry facts, source refs, task context.
Forbidden context	HTTP client details.

Dossier mapper

Allowed context	Sanitized execution metadata and output snapshot.
Forbidden context	Runtime side effects or scoring changes.

Evaluation

Allowed context	Persisted dossier/output and baseline fixture.
Forbidden context	Live provider calls.

20. Extension Points

New source connector

Extension point	config/connectors, connector profile registry, source provider port/adapter.
Required validation	Connector profile tests, preflight, source card validation.

Planner capability change

Extension point	source card compiler, planning input, plan acceptance.
Required validation	Planner/validator fake tests, no-secret trace tests.

Search expansion strategy

Extension point	RadarSearchExpansionService.
Required validation	Unit tests for query families, policy filtering, caps, dedupe.

Registry lookup terms

Extension point	RegistryLookupTermGenerator.
Required validation	Unit tests for aliases, Russian/English/legal-form terms, identifiers, placeholders.

Extraction schema repair

Extension point	extraction contract and OpenRouter provider recovery path.
Required validation	Malformed-output fixtures, retry/backup budget tests.

Entity resolution semantics

Extension point	entity resolution and candidate universe projection.
Required validation	Legal entity vs site/branch/project/asset tests.

Checkpoint policy

Extension point	checkpoint service/action executor.
Required validation	Recorded/fake adaptive pipeline tests.

Signal search behavior

Extension point	retrieval task compiler and staged execution signal phase.
Required validation	Not-searched vs not-observed tests.

Dossier projection

Extension point	API dossier mappers and source lifecycle.
Required validation	Backend API/dossier tests.

Benchmark metrics

Extension point	evaluation module and baseline fixtures.
Required validation	Evaluation unit/report tests.

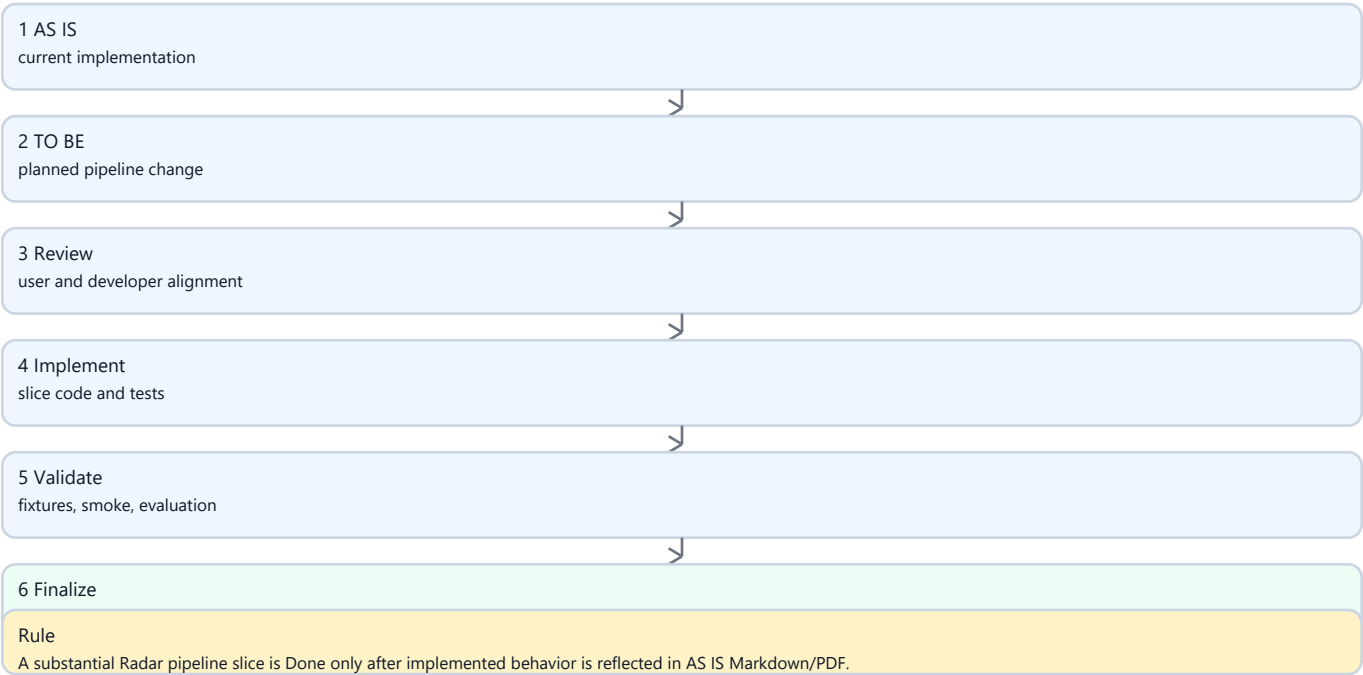
21. Test Map

Behavior	Typical tests
Active definition and runtime wiring	tests/test_radar_preflight.py, tests/test_persisted_live_radar.py
Connector profiles and source cards	tests/test_connector_profiles.py, tests/test_live_icp_radar.py
Source obligations	tests/test_live_icp_radar.py, tests/test_radar_preflight.py
External-call budgets	tests/test_radar_external_call_budget.py
Adaptive checkpoints	tests/test_radar_adaptive_execution.py
Search expansion and lookup terms	tests/test_radar_search_expansion.py, tests/test_live_icp_radar.py
Extraction recovery	tests/test_live_icp_radar.py, tests/test_radar_adaptive_execution.py
Dossier/API projection	tests/test_backend_api.py

Behavior	Typical tests
Benchmark report	tests/test_radar_benchmark.py
Recall/precision evaluation	tests/test_radar_evaluation.py
Backend boundaries	tests/test_backend_architecture_contract.py

22. AS IS / TO BE Maintenance Lifecycle

Figure 6. AS IS / TO BE maintenance cycle



Required maintenance rules:

1. Before a substantial Radar pipeline change, create a TO BE document.
2. Generate or update the slice plan from that TO BE.
3. Implement the slice.
4. Run targeted tests and any required smoke/evaluation flow.
5. Update AS IS with implemented behavior.
6. Regenerate PDF.
7. Mark the roadmap slice Done only after AS IS/PDF are current.

23. Known Gaps

- This document is descriptive. It does not replace tests, dossier, trace, or benchmark reports.
- The AS IS document can drift if future slices skip the maintenance rule; the documentation contract test and slice acceptance criteria exist to reduce that risk.
- The PDF rendering command is intentionally project-local and may need tooling adjustment if the repository moves to another documentation stack.